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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026

(for 2024 Admission Regular Examinations, October 2025)

Third Semester

Discipline Specific Core Course (DSC) – MG3DSCCOM201 – QUANTITATIVE TECHNIQUES FOR BUSINESS

(2024 Admission Onwards)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any 12 questions

Each question carries 1 marks

1. Which branch of statistics is useful for summarizing sales data of a shop? [A] / [CO1]

Descriptive Statistics

2. If the maximum value in a series is 25 and its range is 15, the minimum value of the series is [A] / [CO2]

Range = Maximum − Minimum → 15 = 25 − Minimum → Minimum = 10.

3. If there are three items, what do you call the cube root of the product of those three items?

[U] / [CO1] - Geometric Mean (GM)

4. Which measure of dispersion ignores the signs of deviations? [U] / [CO2]

Mean Deviation

5. What does kurtosis measure in a data set? [U] / [CO3]

Kurtosis measures the degree of peakedness (or flatness) of a frequency distribution as compared to a normal distribution

6. What do you call the reciprocal of the arithmetic average of the reciprocal of the values of a series? [U] / [CO1]

Harmonic Mean.

7. The Coefficient of variation of two series are 58% and 69%. Their Standard Deviations are 21.2 and 15.6. What are their Arithmetic Mean? [A] / [CO2]

CV = (SD / Mean) × 100 → Mean = (SD / CV) × 100

Series 1: Mean = (21.2 / 58) × 100 = 36.55

Series 2: Mean = (15.6 / 69) × 100 = 22.61

8. How is the transpose of a matrix denoted? [K] / [CO4]

It is denoted by A' or A^T (for a matrix A).

9. What is a stub in a table? [U] / [CO5]

The stub is the left-hand column of a statistical table which contains the headings/titles describing the rows of the table.

10. Karl Pearson's coefficient of skewness is 0.32. Its SD is 6.5 and mean is 29.6. Find mode.

[A] / [CO3]

Sk = (Mean − Mode) / SD → 0.32 = (29.6 − Mode) / 6.5

Mode = 29.6 − (0.32 × 6.5) = 29.6 − 2.08 = **27.52**.

11. Consider the data: 2,4,6,8. The Mean Deviation about Mean is **[A] / [CO2]**

Mean = (2+4+6+8)/4 = 5. Absolute deviations: |2−5|=3, |4−5|=1, |6−5|=1, |8−5|=3; Sum = 8.

Mean Deviation = 8 / 4 = 2.

12. In matrices, If A= [-6 -6], B= [-1 -2], find A-B. **[A] / [CO4]**

A − B = [−6−(−1) −6−(−2)] = [−5 −4].

13. A square matrix with 1s on the diagonal and 0s elsewhere is called ..... **[K] / [CO4]**

Identity Matrix (Unit Matrix).

14. Which type of data were collected through questionnaire? **[U] / [CO5]**

Primary data.

[1x12 = 12]

Part B

Short Answer/ Problem Type Questions

Answer any 4 questions

Each question carries 7 marks

15. What are the considerations for choice of an average? Comment. **[U] / [CO1]**

The choice of a suitable average for a given set of data depends on the following considerations:

- Nature of the data — whether qualitative or quantitative, and the level of measurement (nominal, ordinal, interval, ratio).
- Purpose of the investigation — what the average is intended to represent or be used for.
- Type of distribution — whether the data is symmetrical or skewed, with or without extreme values/outliers.
- Existence of open-end classes — median or mode may be preferred over mean when class intervals are open-ended.
- Suitability for further algebraic treatment — arithmetic mean is preferred where further computations (like combined mean, standard deviation) are needed.
- Simplicity and ease of computation and understanding, especially for non-technical audiences.
- Sampling stability — arithmetic mean is the most stable average from sample to sample.

least sampling fluctuation; however, median or mode may be more appropriate for skewed data, qualitative data, or open-end distributions.

Consider any other valid points and answering styles)

16. Calculate Mean Deviation from arithmetic mean of the given series relating to monthly wages.

Wages (Rs.): 21,23,25,28,30,32,38,39,46,48. **[A] / [CO2]**

N = 10; Σx = 21+23+25+28+30+32+38+39+46+48 = 330

Mean ($\bar{x}$) = 330 / 10 = 33 (2 marks)

Wages (x)	**$\lvert x - \bar{x} \rvert = \lvert x - 33 \rvert$**
21	12
23	10
25	8
28	5
30	3
32	1
38	5
39	6
46	13
48	15
Total (N = 10)	**$\Sigma\lvert x - \bar{x} \rvert = 78$**

Mean Deviation from Mean = $\Sigma|x - \bar{x}| / N = 78 / 10 =$ **7.8 (Rs.) – 5 marks**

17. Given $\mu_2=9$, $\mu_3=0$, $\mu_4=200$, analyze skewness and kurtosis. [A] / [CO3]

Skewness (3.5 marks with interpretation)

$\beta_1 = \mu_3^2 / \mu_2^3 = 0^2 / 9^3 = 0 / 729 = 0$

Since $\beta_1 = 0$ ($\gamma_1 = 0$), the distribution is symmetrical — there is no skewness.

Kurtosis (3.5 marks with interpretation)

$\beta_2 = \mu_4 / \mu_2^2 = 200 / 9^2 = 200 / 81 = 2.47$ (approx.)

Since $\beta_2 = 2.47$ is less than 3 ($\gamma_2 = \beta_2 - 3 = -0.53$, negative), the distribution is platykurtic

18. ABC Trading Company has two product lines: Product A and Product B. The company operates in two markets: Urban and Rural. The profit margins (in percentage) and market share data are organized as follows: The profit margin matrix P is given by: Urban market: Product A has 15% profit margin, Product B has 20% profit margin Rural market: Product A has 12% profit margin, Product B has 18% profit margin. Set up the profit margin matrix P as a 2×2 matrix where rows represent markets (Urban, Rural) and columns represent products (A, B). Also find the adjoint of matrix P.

[A] / [CO4]

Profit Margin Matrix P (rows: Urban, Rural; columns: Product A, Product B): - 3 marks

$$\mathbf{P} = \begin{bmatrix} 15 & 20 \\ 12 & 18 \end{bmatrix}$$

For a 2×2 matrix P = [a b; c d], adjoint(P) = [d −b; −c a]. – 4 marks

$$\mathbf{Adjoint(P)} = \begin{bmatrix} 18 & -20 \\ -12 & 15 \end{bmatrix}$$

19. Distinguish between Census Method and Sampling Method. **[U] / [CO5]**

- Coverage: Census covers every unit of the population; Sampling covers only a representative part (sample) of the population.
- Cost: Census is more expensive as it covers the entire population; Sampling is more economical.
- Time: Census takes a longer time to complete; Sampling is quicker.
- Accuracy: Census, in principle, gives more accurate/complete results (subject to non-sampling errors); Sampling results are subject to sampling error but can be highly accurate with proper design.
- Suitability: Census is suitable for small, homogeneous, or when data on every unit is legally required (e.g., population census); Sampling is suitable for large populations and destructive testing.
- Manpower/Resources: Census requires a larger organisation and manpower; Sampling requires comparatively fewer resources.

(**Consider any other valid points and answering styles)**

20. The analysis of the results of a budget survey of 150 families gave an average monthly expenditure of Rs. 120 on food items. After the analysis was completed, it was noted that the figure recorded for one household was wrongly taken as Rs. 15 instead of Rs. 105. Determine the correct value of the average expenditure. **[A] / [CO1]**

Incorrect total expenditure = Mean × N = 120 × 150 = Rs. 18,000

Correct total expenditure = Incorrect total − Wrong value + Correct value

18,000 − 15 + 105 = Rs. 18,090

Correct Mean = 18,090 / 150 = Rs. **120.60**

(7x4 = 28)

Part C

Essay/ Problem Type Questions

Answer any 2 questions

Each question carries 15 marks

21. Calculate Median and Mode from the following data: **[A] / [CO1]**

Class	**0-7**	**7-14**	**14-21**	**21-28**	**28-35**	**35-42**
Frequency	**7**	**11**	**24**	**19**	**12**	**9**

Median (7.5 marks)

Class	**0–7**	**7–14**	**14–21**	**21–28**	**28–35**	**35–42**
Frequency (f)	7	11	**24**	19	12	9
Cumulative Frequency (cf)	7	18	**42**	61	73	82

N = Σf = 7+11+24+19+12+9 = 82

N/2 = 82/2 = 41

The cumulative frequency just exceeding 41 is 42, so the median class is 14–21.

Median = L + [(N/2 − cf) / f] × h

Median = 14 + [(41 − 18) / 24] × 7

= 20.71

Mode (7.5 marks)

Highest frequency = 24, so the modal class is 14–21.

L = 14, f_1 = 24, f_0 = 11 (preceding class), f_2 = 19 (succeeding class), h = 7

Mode = L + [($f_1 - f_0$) / ($2f_1 - f_0 - f_2$)] × h = 14 + [(24−11) / (48−11−19)] × 7 = 14 + (13/18) × 7 = 14 + 5.06

= 19.06 or 19

22. The distribution of marks are given below. Compute the Mean, Standard Deviation and Coefficient of Variation. [A] / [CO2]

Marks	No. of Students
0–20	5
20–40	15
40–60	20
60–80	6
80–100	4
Total	**50**

The student can follow any method-

Using the step-deviation method, with assumed mean A = 50, class width h = 20, and

d′ = (x − A) / h:

Marks	0–20	20–40	40–60	60–80	80–100	Total
Mid-value (x)	10	30	50	70	90	
Frequency (f)	5	15	20	6	4	50
d′ = (x−50)/20	-2	-1	0	1	2	
f d′	-10	-15	0	6	8	-11
f d′²	20	15	0	6	16	57

Mean

Mean = A + (Σfd′ / N) × h = 50 + (−11/50) × 20 = 50 − 4.4 = **45.6 marks (6 marks)**

Standard Deviation (6 amrks)

SD = h × √[(Σfd′²/N) − (Σfd′/N)²]

SD = 20 × √[(57/50) − (−11/50)²] = 20 × √[1.14 − 0.0484] = 20 × √1.0916 = 20 × 1.0448 = **20.90 marks** (approx.)

Coefficient of Variation (3 marks)

CV = (SD / Mean) × 100 = (20.90 / 45.6) × 100 = **45.83%**

23. XYZ Retail Company sells three product categories (Electronics, Clothing, Furniture) across 4 different stores. The company wants to calculate the total daily sales commission for each store. Sales Matrix S (units sold per store): The units sold by each store are as follows (Electronics, Clothing & Furniture respectively): Store 1: 50, 120, 80 Store 2: 40, 150, 90 Store 3: 60, 100, 110Store 4: 45,130, 95 Commission Matrix C (commission per unit in rupees): The commission for each product category is as follows: (Electronics, Clothing &Furniture respectively): Store 1: 100,50, 200 Store 2: 120, 60, 250 Store 3: 110,55, 80 Store 4: 90, 65, 80

a) Prepare the Sales Matrix S and Commission Matrix C.

b) Calculate the daily sales commission for each store.

c) Which store has the highest total sales commission?

d) Which store has the lowest total sales commission? **[A] / [CO4]**

(a) Sales Matrix S and Commission Matrix C (5 marks)

S =

$$\begin{bmatrix} 50 & 120 & 80 \\ 40 & 150 & 90 \\ 60 & 100 & 110 \\ 45 & 130 & 95 \end{bmatrix}$$

C =

$$\begin{bmatrix} 100 & 50 & 200 \\ 120 & 60 & 250 \\ 110 & 55 & 80 \\ 90 & 65 & 80 \end{bmatrix}$$

(b) Daily sales commission for each store (Σ units sold × commission per unit, category-wise) (6 marks)

Store 1 = (50×100) + (120×50) + (80×200) = 5,000 + 6,000 + 16,000 = Rs. 27,000

Store 2 = (40×120) + (150×60) + (90×250) = 4,800 + 9,000 + 22,500 = Rs. 36,300

Store 3 = (60×110) + (100×55) + (110×80) = 6,600 + 5,500 + 8,800 = Rs. 20,900

Store 4 = (45×90) + (130×65) + (95×80) = 4,050 + 8,450 + 7,600 = Rs. 20,100

(c) Highest total sales commission (2 marks)

Store 2, with Rs. 36,300.

(d) Lowest total sales commission (2marks)

Store 4, with Rs. 20,100. **[15x2 = 30]**